Occlusion-Aware Trajectory Planning With Quantified Risk Constraint for Deadlock Mitigation in Autonomous Driving

Zhan Chen, Weidong Liu, Lu Xiong, Zhuoping Yu, and Chen Tang

***Abstract*— Driving through traffic scenes with occlusions safely, efficiently and comfortably remains a huge challenge for autonomous vehicles (AVs). The omnipresence of blind spots, where emergencies may arise unexpectedly, requires AVs to drive with adequate precautions. However, existing planning algorithms often resort to excessive conservatism that leads to navigation-disrupting deadlocks. To address these challenges, we adopt the concept of phantom obstacles, imparting them with uncertainties. At that point, we propose an integrated lateral and longitudinal trajectory planning method to emulate human-like precautionary driving styles, and adopt stochastic model predictive control (SMPC) to model various uncertainties of obstacles. Furthermore, we analyze deadlock causation mechanisms and present a novel quantified risk assessment method featuring an innovative adaptation of ST projection for occlusion-aware planning. This approach formulates a driving strategy that significantly reduces the likelihood of deadlock, enhancing the AV's ability to navigate effectively in occluded scenarios. Case analysis and statistical research across diverse scenarios, featuring different objects in various typical settings, validate the safety, efficiency and comfort of the proposed planning results. These findings underscore the potential of our approach to advance the capabilities of autonomous vehicles in navigating complex and occluded driving environments.**

***Index Terms*— Autonomous vehicle, trajectory planning, occlusion.**

I. Introduction

AUTONOMOUS vehicles significantly enhance traffic efficiency and safety [1], [2], [3]. However, challenging scenarios with occlusions remain a substantial challenge for autonomous driving. In such scenarios, autonomous vehicles encounter formidable challenges due to various uncertainties, including sensor measurement noises, ambiguity regarding the presence of obstacles in occluded areas, and uncertainties in predicting their motion [4], [5], [6], [7], [8], [9]. Even with the most sophisticated sensors, these uncertainties cannot be perfectly mitigated. Consequently, the planning module must account for these uncertainties and suggest a safe yet efficient trajectory. This concept, known as occlusion-aware motion planning, focuses on generating a robust trajectory capable of addressing inherent risks originating from blind spots [10], [11], [12].

In addressing these challenges, recent studies have predominantly centered around structured road environments, particularly intersections. These studies heavily lean on the utilization of map information [14], [15], [16] or draw insights from external data sources [17], [18]. The prevailing approaches often incorporate relatively straightforward conflict obstacles, primarily vehicles. However in practical scenarios, diverse danger situations may involve a myriad of obstacle types, such as pedestrians emerging from behind buses or cyclists appearing at rural forks. This necessitates an algorithm that exhibits compatibility and universality to effectively manage this spectrum of obstacles.

Previous research on occlusion-aware motion planning has also focused on estimating the probability of other traffic participants within blind spots [16]. Some studies adopt a worst-case scenario approach, assuming that virtual obstacles will always emerge from occluded areas, and plan obstacle avoidance using Time-to-Collision (TTC) indices or set-based methods [19], [20], [21]. While effective in ensuring safety, this approach can be overly conservative, often making AVs fall into the so-called freezing state [22]. Another approach involves determining the likelihood of potential obstacles by sampling based on risk coefficients derived from contextual information and road topology [16], [22], though this method lacks sufficient safety countermeasures. However, sampling results are not reliable all the time. It's not a simply yes-or-no binary problem for existence of potential obstacles. When AVs take risks, they should also quantify the level of risk and remain prepared for the worst-case scenario, while avoiding overly cautious planning outcomes that could hinder efficiency.

Regarding planning outputs, many studies primarily focus on longitudinal speed planning, encompassing acceleration and deceleration strategies, when addressing occluded driving areas [23], [24], [25], [26]. In contrast, envisioning a scenario where human drivers encounter such situations, they instinctively make subtle deviations to the other side of the blind area while appropriately slowing down. This combined

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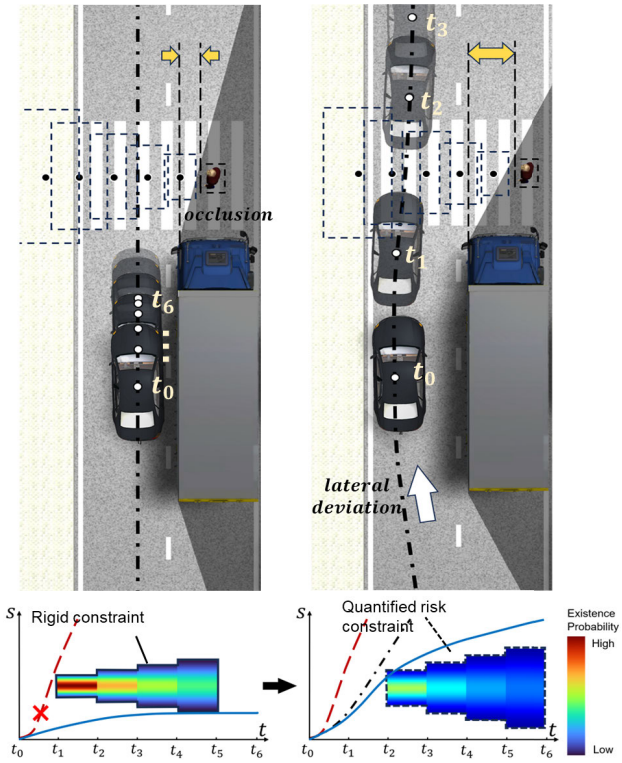

Fig. 1. When AV passes through occluded areas while accounting for the uncertainty diffusion of obstacles, the trajectory planning results that neglect considerations for occlusion (as top-left subfigure), along with the application of rigid constraint on the space-time (ST) projection [13] (as bottom-left subfigure) can lead to conservative planning outcomes. Conversely, an occlusion-aware path introduces lateral deviation, improving the perception of blind spots and expanding the reaction distance for occluded obstacles (as top-right subfigure). The utilization of quantified risk constraints prevents the vehicle from adopting conservative acceleration strategies (as bottom-right subfigure), thereby enhancing the efficiency of occlusion-aware trajectory planning results.

path and speed adjustment not only enhances visibility to concealed traffic participants but also allocates more reaction time for emergency handling. To achieve a driving behavior reminiscent of human drivers, it is imperative to incorporate considerations for both lateral and longitudinal aspects in occlusion-aware trajectory planning.

In this study, the concept of phantom obstacles (PHOs) is adopted as a strategy to address occlusions, and both path and speed are optimized based on Expectation-Maximization (EM) planner [27]. To account for the various uncertainties associated with perceived obstacles (POs) and phantom obstacles, the optimal trajectory planning problem is formulated as stochastic model predictive control (SMPC). These uncertainties are represented as probabilistic constraints, facilitating the effective modelling of risks. To give preference to non-conservative driving strategies, risk and efficiency are balanced through a quantified risk model, which ensures planning results are radical yet exploratory as shown in Fig. 1. This approach aims to avoid deadlock situations while maintaining a primary focus on safety.

The contributions of the proposed method can be summarized as follows:

- We present a versatile occlusion-aware trajectory planning algorithm framework adept at handling diverse traffic scenarios with occlusions. This framework unifies lateral and longitudinal planning and incorporates the concept of PHOs to model uncertainties arising from occluded regions.
- We leverage SMPC due to its inherent probabilistic constraints, facilitating the seamless integration of uncertainties. Based on this, we introduce Quantified Risk Constraints to mitigate deadlock situations and enable safe transitions in highly uncertain environments.
- We conducted extensive experiments in various challenging scenarios, including diverse obstacle types, sequential occluded obstacles, and dynamic occlusions, demonstrating significant improvements in collision avoidance, deadlock resolution, and navigation efficiency compared to other state-of-the-art methods.

The paper is structured as follows. Related works are reviewed in Section II. Section III introduces the combination of EM planner and SMPC to formulate the basic planning problem. Modelling of PHOs for occluded areas are derived in Section III-C. Analyses of the deadlock situation and risk quantification with constraints reformulations are discussed in Section IV-A and B. The simulation results are presented in Section V and conclusion are summarized in Section VI.

II. Related Work

Some methods have been proposed to enable AV to drive through occluded scenes. Notably, the implementation of a vehicle-road cooperative system and inter-vehicle information exchange are aimed at mitigating uncertainties arising from occlusions, thereby ensuring the derivation of speed planning results that prioritize safety and comfort [1], [17], [28], [29]. However, the widespread implementation of these approaches faces challenges, including the requirement for a robust communication network, latency considerations, and the substantial costs associated with infrastructure deployment [18], [30].

In order to uphold a heightened level of safety, it is often assumed that obstacles may potentially emerge from the periphery of the visible area. The TTC index is frequently employed as a cost-effective means of regulating the AV's speed for collision avoidance. However, this method does not provide a robust safety guarantee [19], [31]. As an alternative, the set-based approach is proposed, which considers all possible future states of virtual obstacles, including worst-case scenarios, to formulate reachable sets. The planned trajectory must then ensure non-intersection with these reachable sets to guarantee safety [20]. Despite certain research efforts aimed at refining this approach [14], [15], [25], it tends to results in overly conservative planning outputs.

To mitigate the overcaution associated with planning outcomes, occlusion-aware motion planning can be formulated as a Partially Observable Markov Decision Making Process (POMDP) rather than consistently considering worst-case scenarios [32], [33]. In this approach, uncertainties are integrated into the POMDP model, allowing the inference of the state of

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occluded obstacles through a combination of past and current observations via belief state updates. Building on the POMDP framework, Thornton et al. [34] apply a value sensitive design to vehical speed control, navigating occluded crosswalk. In the work by Schratter et al. [35], Autonomous Emergency Braking (AEB) is formulated using POMDP to address pedestrian jaywalking, though this approach is triggered only when pedestrians are directly perceived, without making inferences about occluded obstacles. In another study by Hubmann et al. [16], changes in the field of view (FOV) are incorporated into a POMDP framework. The probability of existence of potential obstacle is determined by considering traffic density, and sampling is then performed to ascertain the presence of an obstacle in the blind area. Zhang et al. [22] further enhance this approach by incorporating environmental context to derive a more accurate probability of obstacle existence. Nevertheless, these studies heavily rely on sampling results and lack corresponding safety countermeasures. Moreover, the computational cost of POMDP-based methods remains a challenge, despite efforts to enhance solver performance, often involving coarsely discretized action spaces [36].

The learning-based approach achieves human-like driving strategies through the assimilation of extensive driving data, leading to effective planning outcomes tailored to specific scenarios and conflicting entities [37], [38], [39], [40]. However, this method exhibits limitations in terms of generalization and interpretability [41], [42], [43]. In [44], a pioneering Temporal and Global Reasoning Network (ReasonNet) is proposed to incorporate predictions for occluded obstacles using an end-to-end model. Despite its innovation, the network demonstrates inadequacies in the extraction of features related to occluded obstacles, with a notable lack of explicit elucidation. In [45], a sensor model for human drivers is learned for occlusion inference, interpreting observed behaviors of human agents as sensor measurements. Other studies, such as [46] and [47], leverage occupancy flow fields for predictive tasks, showcasing the capability to predict occluded obstacles. Nevertheless, the present precision of this predictive task remains suboptimal, and the network's structural design exhibits deficiencies in feature extraction for occluded obstacles [48].

To model the occluded scenarios, Andersen et al. [49] employ Model Predictive Control (MPC) to enhance the efficiency of navigating past a parked obstacle with occlusion. However, this improvement is achieved solely by enhancing visibility without a direct consideration of the highly probabilistic nature of occluded obstacles, such as their potential sudden appearance. While effective in certain scenarios, this limitation restricts its applicability in more dynamic, uncertain environments. SMPC is a suitable approach for handling objects with significant uncertainties. Brüdigam et al. [50] utilize SMPC in a lane-changing scenario, accounting for uncertainties in sensor measurements and obstacle movements. The approach generates a safe trajectory based on predefined risk parameters. Nevertheless, this study specifically focuses on perceived obstacles within a particular scene, which may not fully capture the challenges of dealing with occluded obstacles or unexpected dynamics in more complex traffic environments.

In summary, current research faces certain limitations, often being tailored to specific scenarios and obstacles, with a heavy reliance on map information to navigate blind spots. Additionally, much of the focus has been on longitudinal velocity adjustments, neglecting the potential benefits of lateral maneuvering. In certain situations, lateral adjustments can significantly enhance visibility, increase the available distance and time to respond to obstacles, and ultimately improve overall efficiency. In our approach, we assumed that PHOs consistently appear at the edge of the visible area during encounters with occlusions. SMPC is employed to optimize the trajectory while considering uncertainties. To mitigate overcaution, which can lead to deadlock situations, we propose a planning strategy that strikes a balance between risk and efficiency, specifically tailored to handle the emergence of obstacles in complex occluded environments. It is important to note that our focus is primarily on motion planning, which necessitates certain simplifying assumptions regarding sensor range and the availability of offline map data.

III. Trajectory Planning With Uncertainty

The conventional EM planner [27] computes trajectory using a combination of dynamic programming (DP) and quadratic programming (QP) methods. In this framework, static obstacles are accounted for during path planning in the Frenét frame, while dynamic obstacles are projected onto the space-time (ST) projection for speed planning. After the introduction of SMPC, the planning process and associated optimization problems need to be reconfigured to address occluded scenarios. The DP and QP problems for path and speed planning in the EM planner, which take into account the uncertainties of POs, will be elaborated in Sections III-A and III-B, respectively. The modifications to the planning framework to account for potential hazards from PHOs in blind spots will be detailed in Section III-C.

A. Decoupled Path and Speed Planning

In trajectory planning, the lane centerline is initially used as the reference line, and perceived static obstacles are projected onto this line. A Lattice sampler is then utilized to compute corresponding cost for each graph edge. These costs take into account multiple objectives, such as obstacle avoidance and path deviation, with respective weights w_i as,

$$\begin{aligned} C(L) &= w_{l1}\sum_{k=1}^{N_l}\sum_{j=1}^{n_{SO}} C_{\text{obs}}(d_k^{\text{obs}(j)}) + w_{l2}\sum_{k=1}^{N_l} l_k^2 \\ &\quad + w_{l3}\sum_{k=1}^{N_l} l_k'^2 + w_{l4}\sum_{k=1}^{N_l} l_k''^2 + w_5\sum_{k=1}^{N_l} l_k'''^2 \\ C_{\text{obs}}(d_k^{\text{obs}(j)}) &= \begin{cases} 1 & d_k^{\text{obs}(j)} \le d_c \\ 0 & d_k^{\text{obs}(j)} > d_c \end{cases} \end{aligned} \tag{1}$$

where, l_k represents lateral deviation at the k-th of N_l points uniformly sampled along the reference line. l_k', l_k'' and l_k''' are related to the yaw angle, the curvature of the path, and the derivative of the curvature, respectively. Additionally, $d_k^{\text{obs}(j)}$ denotes bounding box distance between AV and the j-th static

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obstacle, where the total number of static obstacles is n_{SO}. The parameter d_c is introduced as a buffer to ensure collision avoidance.

A candidate path is then selected through DP by minimizing the overall cost, which determinines the appropriate nudge decisions for static obstacles and establishes a feasible tunnel. Spline-based QP is subsequently adopted to smooth the path, ensuring it remains collision-free and closely aligned with the reference line, as detailed in [13].

B. Probabilistic Constraints for Collision Avoidance

Following path planning, collision avoidance with dynamic obstacles is achieved by generating a speed profile within the ST projection based on Frenét Axis-Aligned Bounding Box (FAABB) [51]. In trajectory planning without considering occlusions, POs are treated as regular objects processed by SMPC to restrict the perturbation caused by uncertainties. The state of each obstacle is defined by its position and velocity in Frenét coordinates as $\boldsymbol{\xi}_k^{\text{obs}} = [s_k^{\text{obs}}, \dot{s}_k^{\text{obs}}, l_k^{\text{obs}}, \dot{l}_k^{\text{obs}}]^\top$, and the dynamics of the obstacle can be modeled as,

$$\boldsymbol{\xi}_{k+1}^{\text{obs}} = \boldsymbol{A}^{\text{obs}}\boldsymbol{\xi}_k^{\text{obs}} + \boldsymbol{B}^{\text{obs}}\boldsymbol{u}_k^{\text{obs}} \tag{2a}$$

$$\boldsymbol{u}_k^{\text{obs}} = \widetilde{\boldsymbol{u}}_k^{\text{obs}} + \omega_k^{\text{obs}}, \hat{\boldsymbol{\xi}}_0^{\text{obs}} = \boldsymbol{\xi}_0^{\text{obs}} + \omega_0^{\text{init}} \tag{2b}$$

with system matrices $\boldsymbol{A}^{\text{obs}}$ and $\boldsymbol{B}^{\text{obs}}$ for the kinematic model. The control input $\boldsymbol{u}_k^{\text{obs}} = [\ddot{s}_k^{\text{obs}}, \ddot{l}_k^{\text{obs}}]^\top$ is composed of a feedback controller $\widetilde{\boldsymbol{u}}_k^{\text{obs}}$ and an independent, identically distributed perturbation ω_k^{obs} on the input. $\widetilde{\boldsymbol{u}}_k^{\text{obs}}$ is generated by tracking the deterministic reference states $\boldsymbol{\xi}_{\text{ref},k}^{\text{obs}}$ with imperfections. The control law is designed as,

$$\widetilde{\boldsymbol{u}}_k^{\text{obs}} = \boldsymbol{K}(\boldsymbol{\xi}_k^{\text{obs}} - \boldsymbol{\xi}_{\text{ref},k}^{\text{obs}}) \tag{3a}$$

$$\boldsymbol{K} = \begin{bmatrix} k_{11} & k_{12} & 0 & 0 \\ 0 & 0 & k_{21} & k_{22} \end{bmatrix} \tag{3b}$$

where, the gain matrix $\boldsymbol{K}$ is calculated from a linear quadratic regulator. ω_k^{obs} is assumed to be a Gaussian distribution with zero mean and covariance matrix $\Sigma_\omega^{\text{obs}}$.

In Eq. (2b), $\hat{\boldsymbol{\xi}}_0^{\text{obs}}$ denotes the believed initial state, assuming the sensor noise $\omega_0^{\text{init}} = [\omega_{s,0}^{\text{init}}, \omega_{\dot{s},0}^{\text{init}}, \omega_{l,0}^{\text{init}}, \omega_{\dot{l},0}^{\text{init}}]^\top$ to be a truncated Gaussian noise as $\omega_0^{\text{init}} \sim \mathcal{N}(0, \Sigma_0^{\text{init}})$ with zero mean and covariance matrix $\Sigma_0^{\text{init}} = diag([\sigma_{s,0}^2, \sigma_{\dot{s},0}^2, \sigma_{l,0}^2, \sigma_{\dot{l},0}^2])$ [50].

Being mapped onto ST projection, both the physical size and the safety redundancy required for uncertainties should be taken into account. Occluded areas pose significant invisible threats, as obstacles may appear suddenly, causing the AV to experience severe disruptions. Therefore, the AV must anticipate potential dangerous situations in advance. Using SMPC, collision checks are incorporated in the form of probabilistic constraints, which ensures that the probability of POs being within the collision region ($\Xi_{\text{safe},k}$) is greater than the predefined risk factor p_{risk},

$$Pr(\boldsymbol{\xi}_k \in \Xi_{\text{safe},k}) \geq p_{\text{risk}} \tag{4}$$

The probabilistic chance constraint is then transformed into a deterministic form for speed optimization. As detailed in [50], covariance propagation for predicted state error $\boldsymbol{e}_k = \hat{\boldsymbol{\xi}}_k^{\text{obs}} - \boldsymbol{\xi}_k^{\text{obs}}$ with Gaussian distribution can be obtained as $\Sigma_{k+1}^e = \boldsymbol{\Phi}\Sigma_k^e\boldsymbol{\Phi}^\top + \boldsymbol{B}^{\text{obs}}\Sigma_\omega^{\text{obs}}\boldsymbol{B}^{\text{obs}\top}$ with $\boldsymbol{\Phi} = \boldsymbol{A}^{\text{obs}} + \boldsymbol{B}^{\text{obs}}\boldsymbol{K}$ and $\Sigma_0^e = \Sigma_0^{\text{init}}$, where $\hat{\boldsymbol{\xi}}_k^{\text{obs}} = [\hat{s}_k^{\text{obs}}, \hat{\dot{s}}_k^{\text{obs}}, \hat{l}_k^{\text{obs}}, \hat{\dot{l}}_k^{\text{obs}}]^\top$ represents the predicted states.

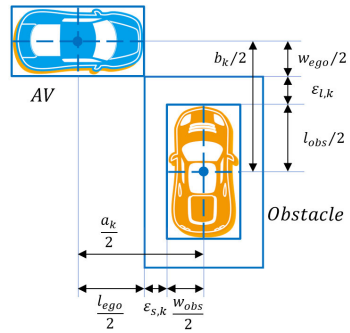

Fig. 2. The longitudinal length and lateral length of the confidence collision region.

Additional safety redundancy in s and l directions can be determined with chi-square distribution of 2 degrees of freedom, (i.e., $\varepsilon_{s,k} = \sigma_{s,k}\sqrt{\chi_2^2(p_{\text{risk}})}$, $\varepsilon_{l,k} = \sigma_{l,k}\sqrt{\chi_2^2(p_{\text{risk}})}$), coupled with geometry of AV (l_{ego}, w_{ego}) and projected length of the dynamic obstacle original physical dimensions in s and l directions ($w_{\text{obs},k}, l_{\text{obs},k}$), as shown in Fig. 3(a), the longitudinal length a_k and lateral length b_k of the confidence collision region are derived as,

$$a_k = l_{\text{ego}} + w_{\text{obs},k} + 2\varepsilon_{s,k} \tag{5a}$$

$$b_k = w_{\text{ego}} + l_{\text{obs},k} + 2\varepsilon_{l,k} \tag{5b}$$

As the prediction progresses, the uncertainty will propagate and accumulate, making the obstacle confidence region gradually expand as shown in Fig. 3(a).

Here a safety threshold for lateral distance (l_{safe}) is considered. Iterate through each POs to identify risky ones if $|\hat{l}_k^{\text{obs}}| - \frac{1}{2}b_k \leq l_{\text{safe}}$, as sketched in Fig. 3(a), and they should be mapped onto the ST projection according to $\hat{s}_k^{\text{obs}}$ and a_k, obtaining POs projection set $\mathbb{O}_{\text{PO}} = \left\{o_{\text{PO}}^{(1)}, o_{\text{PO}}^{(2)}, \ldots, o_{\text{PO}}^{(n_{\text{PO}})}\right\}$, with the superscript (i) denotes the i-th PO obstructing AV with a total number of n_{PO}.

To construct a convex optimization problem, we first determine a convex space on the ST projection through DP. Edge costs with obstacle avoidance, speed tracking and path smoothness are considered as,

$$C(S) = w_{s1}\sum_{k=1}^{N_s} C_{\text{PO}}(s_k) + w_{s2}\sum_{k=1}^{N_s}(s_k' - v_{\text{ref}})^2 + w_{s3}\sum_{k=2}^{N_s} s_k''^2 + w_{s4}\sum_{k=3}^{N_s} s_k'''^2$$

$$C_{\text{PO}}(s_k) = \begin{cases} 1 & (t_k, s_k) \in \mathbb{O}_{\text{PO}}, k = 1, 2, .., N_s \\ 0 & (t_k, s_k) \notin \mathbb{O}_{\text{PO}}, k = 1, 2, .., N_s \end{cases} \tag{6}$$

where, $s_k' = v_k$,$s_k'' = a_k$ and $s_k''' = j_k$ are approximated by the finite difference method.

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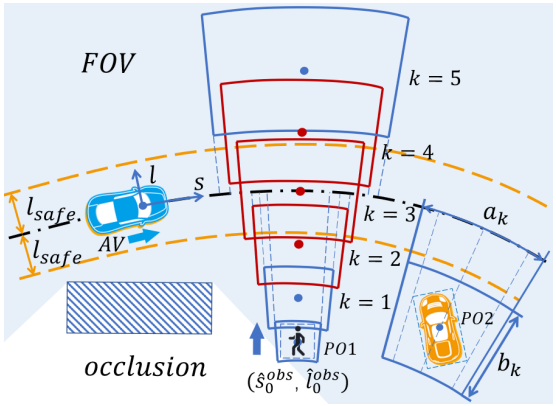
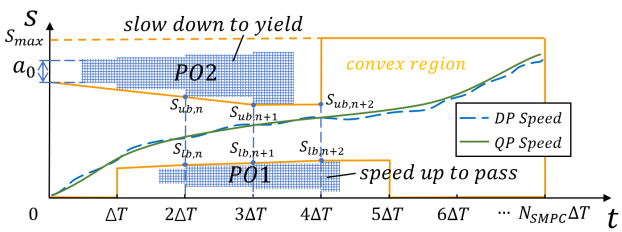


(a)



(b)

Fig. 3. (a) Positional uncertainty of the cut-in pedestrian increases with forward propagation, the red boxes represent the location distributions that pose a threat to AV. (b) ST projection considering probabilistic constraints. Due to the consideration of uncertainty propagation and accumulation, the occupancy block of obstacles on the ST projection will expand over time.

A piecewise linear DP speed profile is then obtained as the guide line, which is illustrated as the blue dashed curve in Fig. 3(b). According to DP results, decision on speed adjustment is made (slow down to yield or speed up to pass the obstacle), and a convex region is generated on ST projection. The probabilistic constraint in the form of Eq. (4) can be transformed into a deterministic manner as,

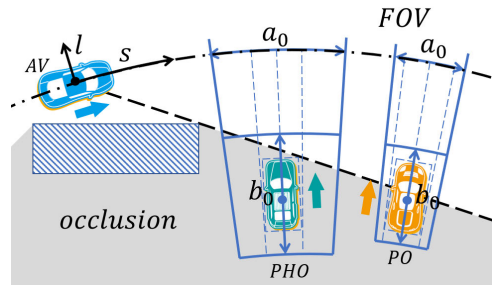
$$s_{lb,k} \leq s_k \leq s_{ub,k} \tag{7}$$

Remark 1: Under the constraint of Eq. (7), optimized speed profile is strictly forbidden to intersect with the area occupied by POs as shown in Fig. 3(b).

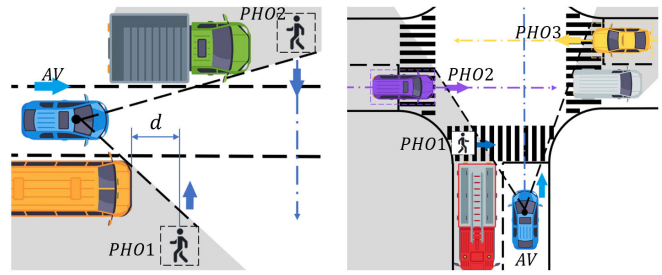
In addition to Eq. (7), constrains on velocity, acceleration and jerk are considered. An optimization problem for speed planning can be formulated as,

$$\begin{aligned} \min_{u_k} \quad & \sum_{k=1}^{N_s} \|\boldsymbol{\xi}_k - \boldsymbol{\xi}_{\text{ref}}\|_{\boldsymbol{Q}}^2 + \sum_{k=0}^{N_s-1} \|\boldsymbol{u}_k\|_{\boldsymbol{R}}^2 \\ s.t. \quad & \boldsymbol{\xi}_{k+1} = g(\boldsymbol{\xi}_k, \boldsymbol{u}_k) \\ & \xi_k \in \boldsymbol{\Xi} \\ & \boldsymbol{u}_k \in \boldsymbol{U} \end{aligned} \tag{8}$$

where $g(\cdot)$ represents longitudinal motion model of AV, $\boldsymbol{\Xi}$ and $\boldsymbol{U}$ denote the feasible sets of system states and control input respectively. $\boldsymbol{Q}$ and $\boldsymbol{R}$ are weight matrices. As sketched in Fig. 3(b), an optimized speed profile can be computed as the green line, which is smoother than DP speed profile.



(a)



(b)

Fig. 4. (a) Both POs and PHOs should be considered when passing through occluded area. (b) Assumed PHOs' locations for constructing the most challenging traffic scenes with occlusions. Dashed lines denotes the edge of occluded area.

C. Trajectory Planning for Blind-Spot

As illustrated in Fig. 4(a), when navigating through occluded areas, both visible POs and invisible PHOs should be considered. Different from POs, PHOs shall be modeled differently due to high virtuality and uncertainty. To construct challenging traffic scenes with occlusion, it is assumed that a PHO suddenly emerges from the specific location at the edge of each occluded area at the next time step, heading straight towards the AV's driveway at top speed, as shown in Fig. 4(b). This assumption prompts the algorithm to proactively consider the most challenging blind spot driving scenarios during the planning process, enabling it to effectively handle the majority of situations.

Remark 2: Based on this assumption, the PHO poses the greatest potential threat to the AV, leaving the shortest reaction time. Avoiding a collision with it ensures safety, even in the presence of multiple occluded dynamic obstacles, as will be demonstrated in the experimental section.

In path planning, it is necessary to keep AV away from the risky areas around PHOs as far as possible by adding distance-related penalty in Eq. (1) as,

$$C(L)' = C(L) + p_{\text{risk}} \cdot w_l \sum_{k=1}^{N_l} \sum_{j=1}^{n_{\text{PHO}}} C_{\text{occ}}(d_k^{\text{PHO}(j)}) \tag{9}$$

where C_{occ} is defined as a monotonically decreasing function while $d_k^{\text{PHO}(j)}$ is the distance between AV and the j-th PHO. The predefined risk factor $p_{\text{risk}} \in [0, 1]$ is the same as the one in Eq. (4) to affect the degree of lateral deviation.

Remark 3: Some approaches for determining risk factor have been proposed in prior works, such as in [22] and [52].

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However, since this is not the primary focus of our study, we will not address it in detail here.

In speed planning, the modeling of PHOs should differ from that of POs. $\hat{\boldsymbol{\xi}}_0^{\text{obs}}$ in Eq. (2b) represents the measured initial state of PO, while subjective initial state of PHO. Correspondingly, ω_0^{init} represents the sensor error for PO and initial prediction error for PHO. It should be noted that the value of $\Sigma_\omega^{\text{obs}}$ and Σ_0^{init} for PHO should be larger than those for PO, in order to introduce greater uncertainty and account for the worst-case scenario.

For POs, it is assumed that they maintain a constant velocity and heading throughout the prediction horizon, represented as $\boldsymbol{\xi}_{\text{ref},k}^{\text{obs}}, k \in \{0, 1, \ldots, N_s\}$. In contrast, the reference state for PHOs is constructed as previously described and shown in Fig. 4. However, due to the highly unpredictable nature of PHO movement in terms of speed and direction, it is assumed that PHOs do not adhere to a specific reference position or velocity in s-direction. Specifically, in Eq. (3b), the parameters $k_{11} = 0$ and $k_{12} = 0$ are set for PHOs.

After obtaining the PHOs projection set $\mathbb{O}_{\text{PHO}} = \left\{o_{\text{PHO}}^{(1)}, o_{\text{PHO}}^{(2)}, \ldots, o_{\text{PHO}}^{(n_{\text{PHO}})}\right\}$, the same algorithm outlined in Section B is applied to derive the planning results.

IV. Occlusion-Aware Path Planning With Quantified Risk

A. The Deadlock Problem

Addressing the uncertainties associated with PHOs and POs by imposing stringent constraints, as defined in Equation (7), leads to safer but more conservative planning outcomes. This cautious approach, however, increases the likelihood of the AV entering a deadlock state. In such situations, the AV may come to a standstill upon approaching occluded areas, unable to proceed further.

To better understand the cause of this freezing state, we analyze it from the perspective of the ST projection. As sketched in Fig. 5(b), it can be observed that the initial projected position of the PHO (t_{start}, s_{start}) on ST plot reveals a specific trend. In s-direction, s_{start} undergoes a process of decreasing change as AV approaches the occlusion and the PHO. In t-direction, as depicted in Fig. 5(a) (assuming the path is a straight line for simplicity), we can get,

$$t_{start} \approx [\frac{d}{B(t)} \cdot A + A - l_{\text{safe}}] \cdot \frac{1}{v_{\text{PHO}}}$$
$$\dot{t}_{start} \approx \frac{v_{\text{AV}}(t)}{B(t)^2} \cdot \frac{Ad}{v_{\text{PHO}}} \tag{10}$$

point C represents the primary corner of the obstacle causing occlusion. The parameter d represents the longitudinal distance between PHO and the corner C, while A denotes the lateral distance between the AV's path and the corner C. They are treated as constants for simplicity. $B(t)$ refers to the longitudinal distance between the AV and the corner C.

To prevent collisions with the PHO and successfully navigate through occlusions, a decisive transition from yielding to preferential passing becomes necessary. This transition is reflected in the ST projection, where the PHO-occupied block undergoes the described movement. This change is depicted

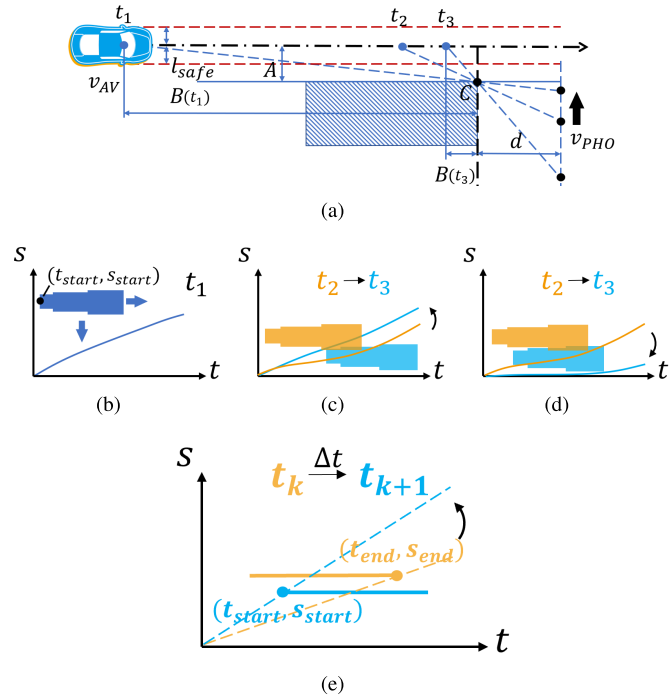

Fig. 5. (a) The sketch to show the relative location of AV and PHO. (b) General movement trend of PHO occupying block. (c) Decision transition success. (d) Decision transition failure. (e)The simplified diagram of speed decision transitions on the ST projection.

in Fig. 5(c), the speed profile shifts from nudging the block from below to nudging it from above. Conversely, failure in making this decision transition, however, leads to a deadlock, as illustrated in Fig. 5(d).

As the AV approaches the occluded area, the projection area of the PHO on the ST projection may be in close proximity to the origin. In such instances, a decision transition entails a significant change in the slope of the speed profile curve. Consequently, the AV is compelled to decelerate, come to a stop, and maintain yielding due to dynamic constraints. Nevertheless, if the AV remains stationary, the assumptions regarding the PHO persist, causing the ST projection to become static, which leads to a freezing state. Simulation analyses of this deadlock situation will be presented in Section V.

In general, the planning of speed profiles on the ST projection is subject to the constraints imposed by vehicle dynamics and the imperative to ensure passenger comfort, specifically with $a \leq a_{max}$. To better illustrate the conditions leading to deadlock, we systematically simplified the ST projection at the moment of decision transition, as illustrated in Fig. 5(e). In this context, we assume the PHO to be a point mass, neglecting the diffusion of uncertainty. This renders the PHO-occupied block as a linear segment. The speed profile is assumed to be a straight line passing through the origin of the ST projection. At time t_k, the decision is to yield to the PHO, and the speed profile intersects the rear end of the PHO's occupying block at (t_{end}, s_{end}). Following a time interval Δt, at time t_{k+1}, the speed decision shifts to prioritize passing, and the speed profile intersects the front end of the PHO-occupied block at (t_{start}, s_{start}). This transition results in an acceleration

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derived as,

$$a = (\frac{s_{start}}{t_{start}} - \frac{s_{end}}{t_{end}})/\Delta t \leq a_{max} \tag{11}$$

Due to the relationship between s_{end} and s_{start}, expressed as $s_{start} = s_{end} - s_{end}/t_{end} \cdot \Delta t$, with Eq. 11, we can get,

$$a = \frac{1 - (t_{start} + \Delta t)/t_{end}}{t_{start}} \cdot s_{end}/\Delta t \leq a_{max} \tag{12}$$

namely, in cases where the specified equation remains unsatisfied, the transition in speed decision cannot occur, necessitating the retention of the decision to decelerate and yield.

In an intuitive sense, to facilitate the satisfaction of the inequality Eq. 12, we aim for a larger value of t_{start}, and smaller values for t_{end} and s_{end}. Correspondingly, based on Eq. (10), we can reach the same conclusion. When AV draws near to occluded area, $B(t)$ is very small, and t_{start} will increase continuously as long as AV moves at an even tiny speed without stopping ($v_{\text{AV}}(t) > 0$). It is to the benefit of the decision transition with smaller acceleration (the slope change of the speed profile). In other words, the process of accessing information and increasing visibility is efficient when AV is close to occlusion. From the simple sketch in Fig. 5(a), compare the process from t_1 to t_2 with that from t_2 to t_3, the latter costs much less distance to obtain much more information in occluded area. Therefore, keeping AV moving is crucial and significant to speed decision transition. In fact, the lateral deviation in path planning also increases the value of t_{start} and visibility, which is in line with our intuition as human driver [53]. Thus, we propose that the solution to deadlock is to incentivize AV to continuously explore blind spots to obtain information and promote the decision-making from yielding to acceleration while controlling the degree of risk and keeping small fluctuation of speed profile to avoid drastic acceleration and deceleration. We propose a method incorporating quantified risk constraints, which will be elaborated in the following section.

B. Probabilistic Constraints With Quantified Risk

Given PHOs with high virtuality and additional probabilistic safety redundancy, treating PHOs as rigid POs by imposing strict constraints may be overly conservative. A more appropriate approach is to model them as the risk probability distribution on the ST projection, i.e., areas occupied by PHOs are not completely impenetrable; rather, penetrating these regions involves balancing efficiency with acceptable risk levels.

To accurately quantify the risk posed by PHO at each moment and location, we leverage the probabilistic information regarding the confidence collision regions of the PHOs. This allows us to evaluate the likelihood of potential collisions dynamically. To model this, we employ the probability density function (normpdf) of the normal distribution and the cumulative distribution function (normcdf). These functions provide a mathematical framework for assessing the risk distribution of PHOs over time and space, as formulated below,

$$f_P(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp[\frac{-(x-\mu)^2}{2\sigma^2}] \tag{13a}$$

$$F_C(x; \mu, \sigma) = \int_{-\infty}^{x} f_P(t; \mu, \sigma) dt \tag{13b}$$

Since the locations of PHOs in both s and l directions follow a truncated Gaussian distribution respectively, with $\varepsilon_{s,k} = \sigma_{s,k}\sqrt{\chi_2^2(p_{\text{risk}})}$ and $\varepsilon_{l,k} = \sigma_{l,k}\sqrt{\chi_2^2(p_{\text{risk}})}$, the probability that s-value and l-value of j-th PHO geometric center $\hat{s}_k^{\text{PHO}(j)}$ and $\hat{l}_k^{\text{PHO}(j)}$ (abbreviated as $\hat{s}_k^{(j)}$ and $\hat{l}_k^{(j)}$) lie within respective interval $[\hat{s}_k^{(j)} - \varepsilon_{s,k}^{(j)}, \hat{s}_k^{(j)} + \varepsilon_{s,k}^{(j)}]$ and $[\hat{l}_k^{(j)} - \varepsilon_{l,k}^{(j)}, \hat{l}_k^{(j)} + \varepsilon_{l,k}^{(j)}]$ is defined as $P_{total}(p_{\text{risk}}) = F_C(\sqrt{\chi_2^2(p_{\text{risk}})}; 0, 1) - F_C(\sqrt{\chi_2^2(p_{\text{risk}})}; 0, -1)$. Then Eq. (6) is reconstructed as,

$$C(S)' = C(S) + w_s \sum_{k=1}^{N_s} \sum_{j=1}^{n_{\text{PHO}}} C_{\text{PHO}}(s_k, p_{\text{risk}}, \hat{s}_k^{(j)}, \hat{l}_k^{(j)}) \tag{14}$$

where, the POs projection set remains strictly prohibited from intersecting.

To calculate C_{PHO}, we connect (t_{k-1}, s_{k-1}) and (t_k, s_k), and divide it into n segments. The risk cost of each node is accumulated. Considering the physical sizes of both the AV and PHO, the risk probability distribution of PHO is illustrated in Fig. 6. This distribution can be viewed as the synthesis of the respective risk levels posed by the PHO in two directions. In s direction, the colliding probability varies with longitudinal position of AV (divided into risk of yielding and risk of passing with priority). In l direction, the degree of threat that PHO poses to AV evolves over time (distinguished between an incipient threat and one that is about to end). In the following, $F_C(\mathcal{X}^{(j)}; \hat{s}_k^{(j)}, \sigma_{s,k})$ and $F_C(\mathcal{X}^{(j)}; \hat{l}_k^{(j)}, \sigma_{l,k})$ are abbreviated as $F_{C,k}^{s(j)}(\mathcal{X})$ and $F_{C,k}^{l(j)}(\mathcal{X})$ respectively.

Quantified risk for PHOs is formulated as,

$$\begin{aligned} &C_{\text{PHO}}(s_k, p_{\text{risk}}, \hat{s}_k^{(j)}, \hat{l}_k^{(j)}) \\ &= w_{\text{span}} \cdot C_{\text{PHO}}^l(\hat{l}_k^{(j)}, p_{\text{risk}}) \cdot \sum_{i=1}^{n} e^{v(i)} \cdot C_{\text{PHO}}^s(s^{(i)}, \hat{s}_k^{(j)}, p_{\text{risk}}) \end{aligned} \tag{15}$$

where, C_{PHO}^s and C_{PHO}^l represent quantified risk in s and l direction respectively. The term $e^{v(i)}$ reflects that as the speed of the AV increases, the potential danger also escalates. The parameter w_{span} is introduced to penalize the hazardous behavior of spanning the PHO occupied area. We consider this spanning behavior to carry extremely high risk, and if it occurs, it will be penalized as equivalent to a collision, as (16)–(18), shown at the bottom of the next page.

After quantifying the risks associated with PHOs and obtaining the initial profile, the original 2D planning space of the ST projection is transformed into a more comprehensive 3D planning space, where the additional dimension explicitly quantifies risk information. A feasible tunnel is generated accordingly, ensuring that the feasible region does not intrude into areas occupied by POs, as aforementioned. However, as occupation blocks of PHOs are dynamically updated based on both spatial and temporal uncertainties, adjustments concerning PHOs are handled differently, as detailed in Algorithm 1.

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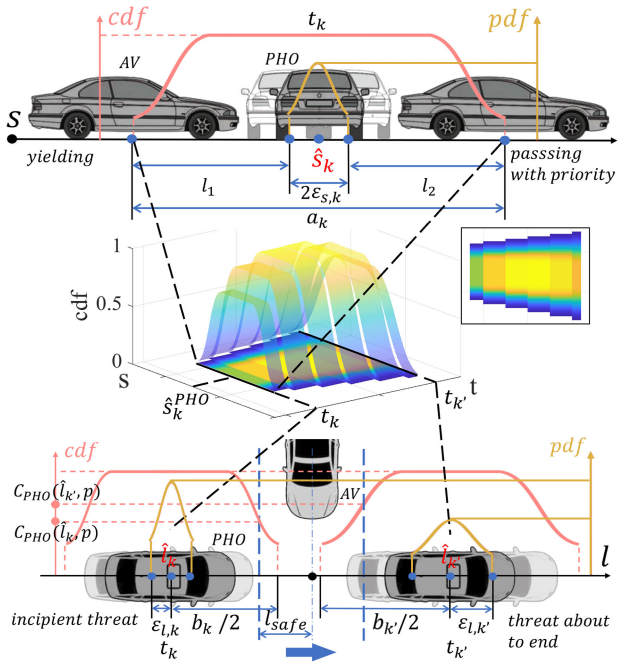

Fig. 6. The illustrated risk probability distribution of PHO on ST projection and disassembly in s and l directions. The derivation of risk cdf is based on risk pdf and relative position of AV and PHO, as detailed in Eq. (16) and (17).

Algorithm 1 Update Of Convex Region Against PHOs

for *each* $o_{PHO}^{(i)} \in \mathbb{O}_{PHO}$ **do**
  if *DP speed profile* $\cap o_{PHO}^{(i)} \neq \emptyset$ **then**
    find the intersection starting point (t_{in}, s_{in}) and corresponding $\hat{s}_{in}^{\mathrm{PHO}(i)}$, $j = 0$;
    while $(t_{in+j}, s_{in+j}) \in o_{PHO}^{(i)}$ **do**
      if $s_{in} \leq \hat{s}_{in}^{PHO(i)}$ **then**
        $s_{ub,in+j} = \min(s_{ub,in+j}, s_{in+j})$;
      else
        $s_{lb,in+j} = \max(s_{lb,in+j}, s_{in+j})$;
      end
      $j++$;
    end
    handle same as PO for sections outside $o_{\mathrm{PHO}}^{(i)}$;
  else
    handle same as PO for $o_{\mathrm{PHO}}^{(i)}$;
  end
end
return updated convex region

Updated convex region relaxes part of the constraints caused by PHOs as shown in Fig. 11, and a less conservative longitudinal planning results can be obtained by solving the optimization problem, achieving the purpose that freezing state is avoided and the risk to a certain degree AV should take is clearly known.

V. Experimental Validation

In this section, we present a series of complex traffic scenarios designed to simulate occlusion challenges, based on real-world situations as illustrated in Fig. 7. Scenario 1 features two trucks parked along the roadside, creating potential hazards due to blind spots behind either vehicle. Initially occluded obstacles are poised to emerge from behind the second truck as the AV approaches, raising significant safety concerns. Scenario 2 depicts a bus stop with a stationary bus, a typical and potentially hazardous situation. As the AV approaches, occluded obstacles are on the verge of crossing the road from the blind spot behind the bus, further amplifying the inherent risks of the scenario. Scenario 3 considers an unsignalized intersection, where a preset obstacle vehicle emerging from a blind spot is occluded by a truck, introducing a potential threat to the AV. To demonstrate the general applicability of the proposed method in addressing both static and dynamic occlusions, we conduct simulations with the truck in both parked and moving states. In these scenarios, the AV is tasked with commencing its journey from a consistent starting

$$C_{\mathrm{PHO}}^{s}(s^{(i)}, \hat{s}_k^{(j)}, p_{\mathrm{risk}}) = \begin{cases} \max\{0, \min[1, \dfrac{F_{C,k}^{s(j)}(\hat{s}_k + \varepsilon_{s,k}) - F_{C,k}^{s(j)}(s^{(i)} - l_1)}{P_{total}(p_{\mathrm{risk}})}]\}, \\ \quad if \quad s_{k-1} > \hat{s}_k^{(j)} \qquad \text{(risk of yielding)} \\ \max\{0, \min[1, \dfrac{F_{C,k}^{s(j)}(s^{(i)} + l_2) - F_{C,k}^{s(j)}(\hat{s}_k - \varepsilon_{s,k})}{P_{total}(p_{\mathrm{risk}})}]\}, \\ \quad if \quad s_{k-1} \leq \hat{s}_k^{(j)} \qquad \text{(risk of passing with priority)} \end{cases} \tag{16}$$

$$C_{\mathrm{PHO}}^{l}(\hat{l}_k^{(j)}, p_{\mathrm{risk}}) = \begin{cases} \max\{0, \min[1, \dfrac{F_{C,k}^{l(j)}(\hat{l}_k + \varepsilon_{l,k}) - F_{C,k}^{l(j)}(\varepsilon_{l,k} - b_k/2 - l_{\mathrm{safe}})}{P_{total}(p_{\mathrm{risk}})}]\}, \\ \quad if \quad \hat{l}_k^{(j)} < 0 \qquad \text{(incipient threat)} \\ \max\{0, \min[1, \dfrac{F_{C,k}^{l(j)}(-\varepsilon_{l,k} + b_k/2 + l_{\mathrm{safe}}) - F_{C,k}^{l(j)}(\hat{l}_k - \varepsilon_{l,k})}{P_{total}(p_{\mathrm{risk}})}]\}, \\ \quad if \quad \hat{l}_k^{(j)} \geq 0 \qquad \text{(threat about to end)} \end{cases} \tag{17}$$

$$w_{\mathrm{span}} = \begin{cases} 1, & (s_{k-1} - \hat{s}_k^{\mathrm{obs}}) \cdot (s_k - \hat{s}_k^{\mathrm{obs}}) > 0 \\ w_s 1, & (s_{k-1} - \hat{s}_k^{\mathrm{obs}}) \cdot (s_k - \hat{s}_k^{\mathrm{obs}}) \leq 0 \end{cases} \tag{18}$$

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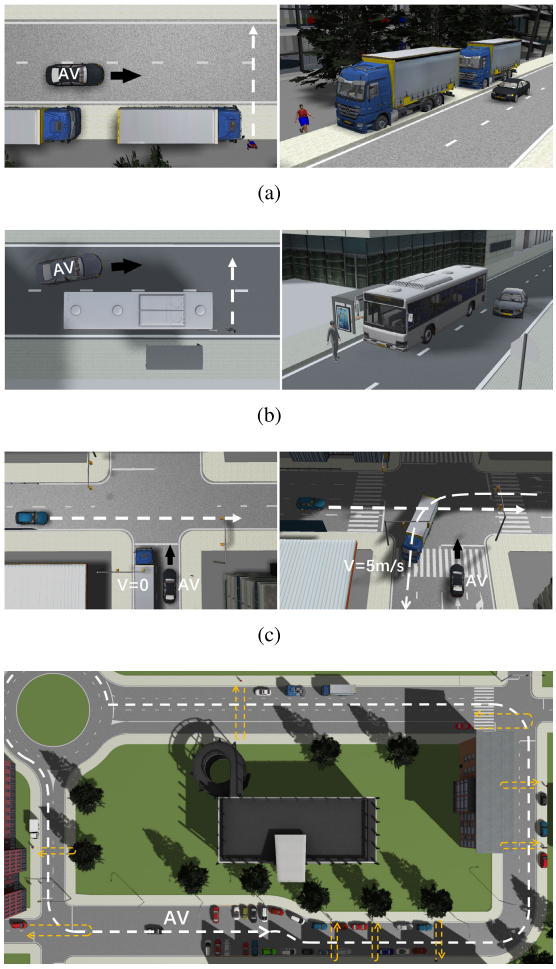

Fig. 7. (a) Scenario 1: a road segment with multiple trucks parked along the side. (b) Scenario 2: a bus station with parked bus. (c) Scenario 3: an intersection with limited visibility caused by the truck under static (left) and moving (right) condition. (d) Scenario 4: a challenging, infinitely looping road filled with blind spots, where multiple pedestrians repeatedly cross the road at occluded locations, and several vehicles are arranged to suddenly emerge from the blind spots.

position in front of the blind spots, traversing through the occlusions, and reaching a designated endpoint.

According to [54], we aim to maximize the criticality of the scenarios as much as possible. As shown in Fig. 8, we conduct experiments with various types of obstacles, including pedestrians, bicyclists, and cars. The longitudinal distance d between the PHO and the corner of the blind spot is either uniformly set to $2m$ for general cases, or calculated based on the centerline of the structured road and crosswalk when such geometry is available. We use TTC as a criterion for scenario construction. We monitor the AV's speed and trigger the cut-in motion of the occluded obstacle from the blind spot when the AV is expected to reach the potential collision zone within 0.5 seconds. Specifically, the average speed v_{ave} is computed to ensure that the obstacle also reaches the collision zone within 0.5 seconds, so that the TTC is 0.5. By adjusting the distance between the initial position of the occluded obstacles and the potential collision zone, we ensure that the v_{ave} of pedestrians, bicyclists, and cars to approximately $2m/s$, $5m/s$,

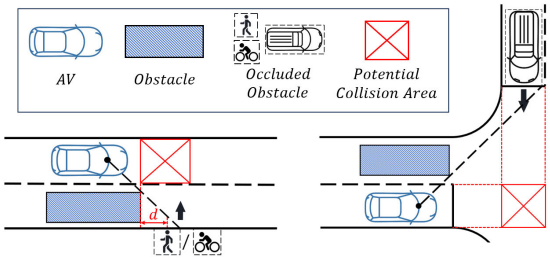

Fig. 8. The illustration of critical scenario setup.

TABLE I

KEY PARAMETERS APPLIED IN TRAJECTORIES PLANNING

Parameter	Value	Parameter	Value
Max.steering angle	$0.4rad$	Max.acceleration	$4m/s^2$
Min.steering angle	$-0.4rad$	Min.acceleration	$-6m/s^2$
Max.curvature	0.25	AV reference speed	$8m/s$
Min.curvature	-0.25	Planning frequency	$10Hz$

and $10m/s$, respectively, which are relatively high speeds for each type of obstacle and pose a safety threat to the AV. Furthermore, to increase experimental uncertainty and realism, we sample the time-varying longitudinal velocity of occluded obstacles from $\text{clip}(\mathcal{N}(v_{\text{ave}}, 1), 0, \infty)$, and sample the noise added to the initial lateral position of the dynamic obstacle from $\text{clip}(\mathcal{N}(0, 1), -d, d)$. This introduces randomness in the interaction positions within the potential collision zone.

Additionally, to evaluate the proposed method's overall performance in more complex traffic situations, we adopted an infinitely looping road as Scenario 4 with numerous perception blind spots caused by other vehicles or buildings. In several of these blind spots, repeatedly pedestrians cross the road, while vehicles suddenly emerge from the occluded areas. This looping setup ensures randomness and diversity in the scenario, where a moving obstacle may appear as either a PO or a PHO, both posing significant safety challenges to the AV.

To assess the effectiveness of the proposed occlusion-aware trajectory planning method, co-simulations are conducted using Simulink, CarSim, and PreScan. To balance computational real-time performance and trajectory planning effectiveness, we select a maximum longitudinal planning distance of 60 meters and a maximum planning time of 6 seconds for path and speed planning, respectively. Table I provides an overview of some other key parameters utilized in these simulations.

A. Effect of Lateral Deviation And Risk Factor

We initially conduct experiments to assess the influence of varying degrees of lateral offset in three distinct scenarios. Notably, the experimental setups exclude occluded obstacles to eliminate the stochastic interference caused by obstacle unpredictability. The experimental outcomes are depicted in Fig. 9. Primarily, higher risk factors result in more conservative trajectory planning outcomes. This observation indicates that elevated risk factors lead to a greater distance from the potentially occluded obstacle when traversing the blind spot, with the decision transition from deceleration to acceleration occurring later. Additionally, the findings highlight that intro-

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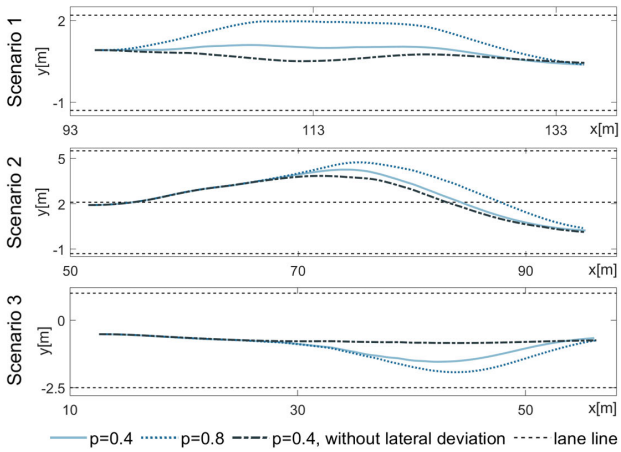
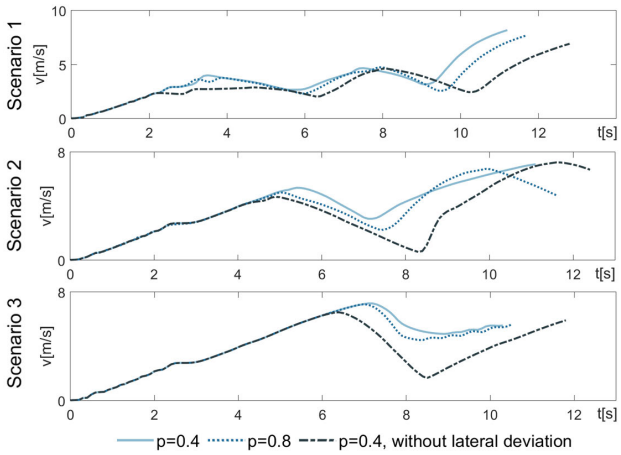


(a)



(b)

Fig. 9. Comparison paths and speeds under different risk factors p and varying degrees of lateral deviation: (a) paths, (b) speeds.

ducing judicious lateral deviation during blind spot traversal, as opposed to scenarios with no lateral offset, facilitates the transition in speed decision-making from deceleration to prioritized passage, thereby significantly enhancing traffic efficiency.

In Scenario 1, an occluded pedestrian is introduced, maintaining a risk factor of $p = 0.4$. To thoroughly assess the impact of lateral deviation on collision avoidance, a series of experiments (200 trials) are conducted under two conditions: with and without lateral deviation. The primary objective of these experiments is to examine the distance between the AV and the pedestrian at the moment when the AV first perceives the initially occluded pedestrian in each scenario. The statistical outcomes of these trials are depicted in Fig. 10. The analysis reveals a notable pattern, where lateral deviation significantly increases the distance at which the AV initially detects the occluded pedestrian. From a practical perspective, this increased distance translates into an expanded reaction space and additional time for both the AV and the obstacle to take necessary precautions. This extended reaction window effectively reduces the probability of a collision by allowing for more proactive safety measures and smoother avoidance maneuvers. Such findings underscore the importance of

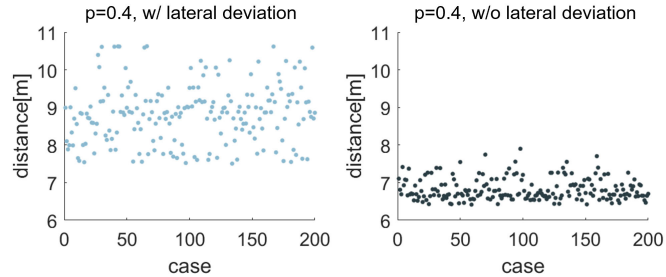


Fig. 10. Statistical experimental results for Scenario 1 under conditions with and without lateral deviation.

incorporating lateral deviation strategies into occlusion-aware planning frameworks, as it substantially enhances the AV's capability to safely navigate environments involving unpredictable obstacles.

B. Leveraging Quantified Risk Constraints for Deadlock Resolution

To validate the effectiveness of quantified risk constraints in overcoming deadlock situations, we conduct the following comparative experiments. These tests aim to comprehensively assess the trajectory planning performance of different methods when the AV approaches a potential freezing state. For this comparison, we select two well-established methods for testing: GO-POMDP [22] and the set-based method [14], both of which serve as representative approaches in this context. In addition, we test the proposed method under two configurations—one using hard constraints and the other utilizing quantified risk constraints. The experiments are conducted in Scenarios 1 and 2. To isolate the impact of blind spot handling on algorithm performance, we temporarily remove all moving obstacles, leaving only the static obstacles that create blind spots. This setup allows for a focused evaluation of each method's ability to prevent deadlock and ensure smooth trajectory planning in the presence of occlusions.

As shown in Fig. 11, the GO-POMDP method, due to its sampling-based characteristics, exhibits significant stopping behavior when approaching blind spots. Although this ensures the avoidance of deadlock, the traffic efficiency is low. Furthermore, the discrete nature of the planning results leads to noticeable fluctuations in the final control output, which compromises driving comfort. On the other hand, the set-based method demonstrates a highly conservative approach, especially in the challenging blind spot scenarios we constructed. When obstacles are large and fully block perception, the set-based method tends to make the most conservative assumptions, which restrict the feasible region and ultimately result in a deadlock, with the AV coming to a complete stop.

For the proposed method, we capture ST projection snapshots at two key moments (A and B as shown in Fig. 11) in both scenarios to illustrate the results. Due to the consideration of PHO uncertainties, the PHO expands its occupied space in the ST projection, squeezing the feasible convex space. This effect is particularly evident in Scenario 1, where continuous blind spots lead to the assumption of continuous PHOs. Under rigid constraints, these conservative assumptions force the

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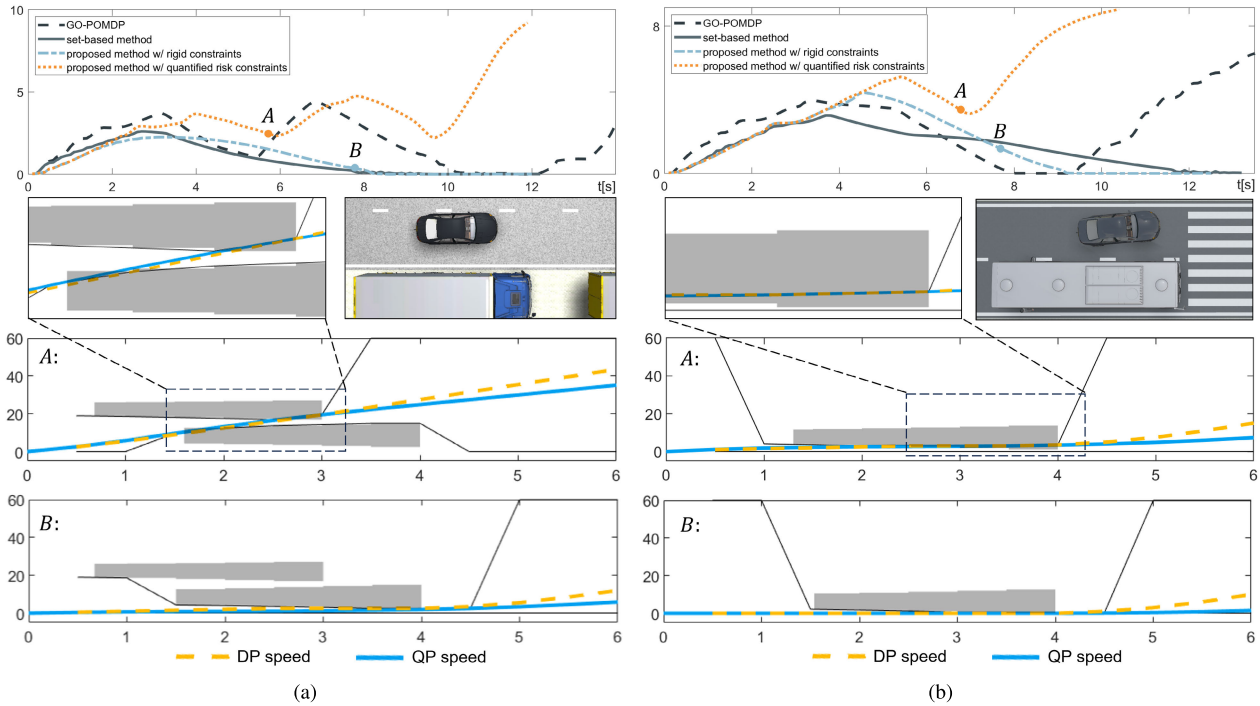

Fig. 11. In the simulations of Scenarios 1 and 2, snapshots of the ST projection at two distinct time points illustrate that the introduction of quantified risk constraints can establish a onward feasible tunnel, effectively resolving deadlock situations. (a) Scenario 1, (b) Scenario 2.

speed curve in the ST projection to pass through the lower region, ultimately resulting in deadlock and a halt in vehicle motion. In contrast, under quantified risk constraints, the speed curve actively opens up convex space in the ST projection in certain extreme cases, maintaining forward progress when the risk is minimal, as shown at moment A in Scenario 1. Even in more challenging conditions, like moment A in Scenario 2, where the speed curve is forced to plan through the lower region due to the absence of convex space above the occupied area, the quantified risk constraint allows the speed curve to maintain its slope, continuing forward and exploring blind spot information to avoid deadlock. Overall, the speed profile generated by the proposed method achieves the highest traffic efficiency compared to the other methods.

C. Universality and Safety

1) Evaluation in Representative Scenarios: To validate the universality and safety of the proposed method, we conduct multiple experiments in four constructed scenarios. These experiments include all essential scene elements, particularly pre-set occluded obstacles within blind spots. As mentioned earlier, these obstacles move at relatively high speeds with fluctuations to ensure randomness, but they are designed to cause collisions with the AV if no precautions are taken. The baseline method selected for comparison is the EM planner [13] (without considering the uncertainty of dynamic obstacles or PHOs). Additionally, we compare the proposed method against GO-POMDP and the set-based method. To ensure a more comprehensive comparison, we enhance the original set-based method [14] by integrating the acceleration-based occupancy modeling proposed in [20], allowing it to account for the motion uncertainty of occluded obstacles. Specifically, we adopt initial velocities of $2m/s$, $5m/s$, and $10m/s$ for pedestrians, bicyclists, and cars, respectively, and set the maximum acceleration a_{max} to $10m/s^2$ following the configuration in [20]. To represent lateral position uncertainty, we add a random offset sampled from the range $[-d, d]$ to the initial lateral position of the obstacle.

We conduct multiple tests in Scenarios 1, 2 and 3. The focus is on the interval during which the AV's speed is influenced by the occlusion until it fully perceive the conditions within the blind zone. A comparative analysis is performed in terms of passage time and collision rates. In Scenarios 1 and 2, to assess the adaptability of the proposed method to different types of obstacles, we test various dynamic obstacle categories, specifically occluded pedestrians and occluded bicyclists. Furthermore, to demonstrate the method's applicability to scenarios involving consecutive occluded obstacles, we set up cases in three scenarios where two obstacles sequentially enter the potential collision zone. As shown in Fig. 13, in Scenarios 1 and 2, the two obstacles are of different types (pedestrian and cyclist), with their entry order randomized. The first obstacle has a TTC of 0.5, while the second obstacle enters the zone 1s later. Simultaneously, to demonstrate the proposed method's universality in both static and dynamic scenarios, we conduct simulations in Scenario 3, where a truck moves dynamically. In this setting, the truck makes a left turn at an intersection, blocking the AV's perception of occluded vehicles, as shown in Fig. 7(c). Each experimental setup is tested 200 trials to ensure robustness.

The experimental results are presented in Fig. 12 and Table II. For the baseline method, the lack of consideration for potential hazards within the blind spot and the absence

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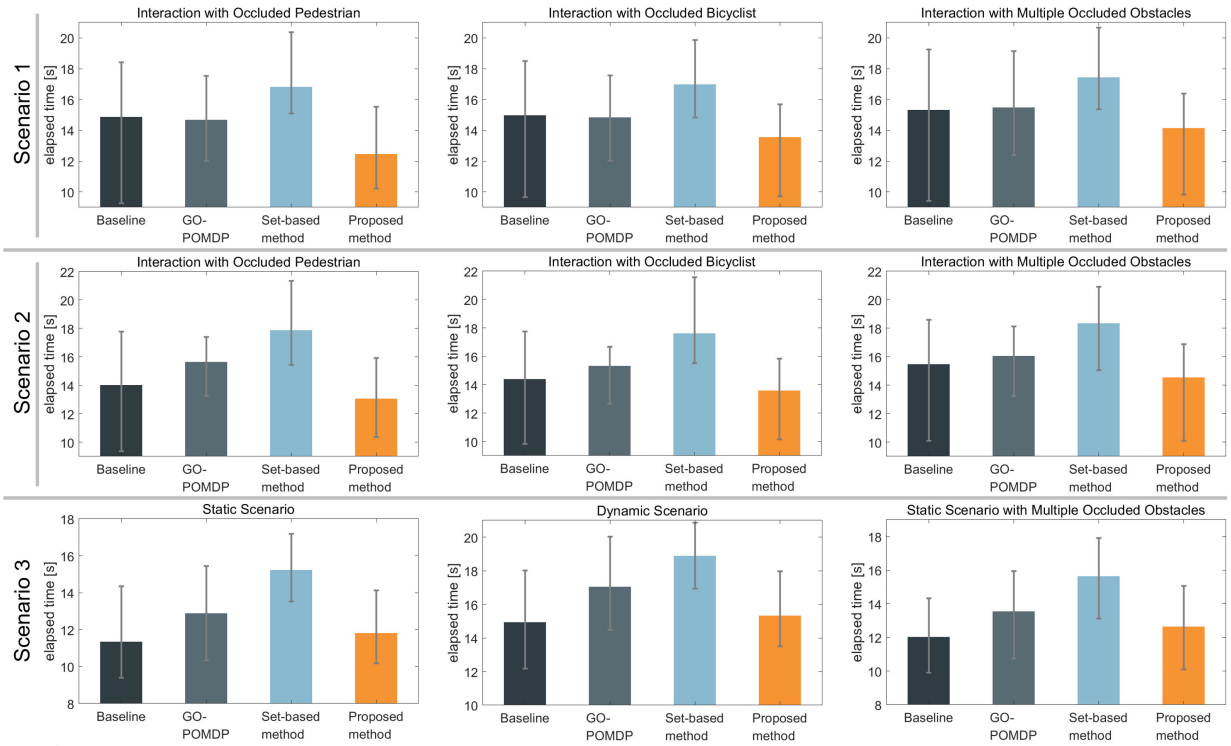

Fig. 12. Barplots of average time consumption for blind spot traversal in three scenarios.

Fig. 13. The testing scenario with multiple occluded obstacles.

TABLE II
COLLISION RATES (%) ACROSS SCENARIOS AND METHODS

Method	Scenario 1		
	Occluded Pedestrian	Occluded Bicyclist	Multiple Occluded Obstacles
Baseline [13]	73.0%	76.5%	80.5%
GO-POMDP [22]	34.5%	29.0%	41.5%
Set-based method [14], [20]	0.0%	0.0%	0.0%
Proposed method	2.0%	3.0%	2.0%
Method	Scenario 2		
	Occluded Pedestrian	Occluded Bicyclist	Multiple Occluded Obstacles
Baseline [13]	58.5%	55.0%	66.5%
GO-POMDP [22]	25.0%	30.5%	37.5%
Set-based method [14], [20]	0.0%	0.0%	0.0%
Proposed method	2.0%	4.0%	5.0%
Method	Scenario 3		
	Static Scenario	Dynamic Scenario	Static Scenario with Multiple Occluded Obstacles
Baseline [13]	77.5%	48.0%	83.0%
GGO-POMDP [22]	42.5%	32.5%	47.0%
Set-based method [14], [20]	0.0%	0.0%	0.0%
Proposed method	3.0%	2.0%	4.5%

of collision avoidance for occluded obstacles result in a significantly higher number of collision cases. Additionally, the variation in passage times across different trials is considerable in most experimental scenarios, showing inconsistency and a lack of robustness in performance. In comparison, the GO-POMDP method, which incorporates an observation and sampling process, exhibits overall longer passage times. However, the collision rate is significantly reduced compared to the baseline, as the method considers safety for potential obstacles within the blind spot. Still, the collision rate remains relatively high due to the method's heavy reliance on the sampling process. Many collisions occur when the method blindly trusts the sampling result (i.e., no obstacles within the blind spot), only to crash into suddenly appearing obstacles. This outcome underscores the necessity of maintaining the assumption of the existence of PHOs within occluded areas to enhance safety. On the other hand, the set-based method effectively reduces the collision rate to a very low level. However, this safety improvement comes at the cost of compromised traffic efficiency. Specifically, the set-based method frequently results in deadlock situations, where the ego vehicle is unable to proceed due to over-conservative risk estimation, despite the actual path being safe and navigable [22]. This issue is particularly prominent in scenarios involving occlusions, where overly cautious assumptions about potential obstacles lead the vehicle to halt indefinitely. Our experiments show that the set-based method results in overall deadlock rates of 93.0%, 87.0%, and 83.5% in Scenario 1, Scenario 2, and the static Scenario 3, respectively (in dynamic Scenario 3, where the truck eventually moves and the blind spot disappears, deadlocks are avoided). In addition, it leads to the longest average elapsed time across all tested scenarios. Such a high deadlock rate and prolonged navigation time are unacceptable for autonomous vehicles aiming to achieve safe yet efficient navigation in real-world driving environments.

In contrast, the proposed method achieves a deadlock rate of 0% in all evaluated scenarios, demonstrating both high efficiency and safety with competitive passage times. Notably,

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the passage times of the proposed method in several scenarios are similar to, or even better than, the baseline method. Furthermore, the proposed method consistently maintains an extremely low probability of collisions, effectively balancing safety and performance. The method also exhibits remarkable adaptability and robustness across a diverse range of obstacle types, including pedestrians, bicyclists, and vehicles, as well as in varying scenarios, from static to dynamic environments. Even in highly complex situations involving sequentially occluded multiple obstacles, the proposed method continues to perform reliably, showcasing significant advantages in both efficiency and safety when compared to alternative approaches. Additionally, the computational efficiency of the proposed method is noteworthy, with the average computation time per planning iteration across all test trials increasing by only 6.8% compared to the baseline. These results underscore the method's capability to handle complex traffic scenarios with occlusions, ensuring smooth and safe navigation even under challenging conditions.

2) Testing in the Integrated Scenario: To evaluate the comprehensive performance and scalability of the proposed algorithm in continuous scenarios, we conduct extensive tests using the constructed Scenario 4. The experiment is designed as an infinite mileage test, where the AV continuously drives on a looping road until either a collision occurs or it falls into a deadlock, at which point the simulation concludes. All dynamic obstacles, which pose varying levels of threat to the AV, move back and forth along predetermined paths, introducing significant uncertainty to the scenario. These obstacles include both POs and occluded obstacles, with the moving agents potentially being either visible or hidden. This setup is specifically designed to evaluate the algorithm's overall performance in managing both direct and potential risks in highly complex scenarios.

In addition to the infinite mileage test, a fixed-time simulation test is also conducted to further analyze the algorithm's scalability and robustness. In this experiment, the AV operates for a fixed simulation duration of three minutes, during which the maximum distance traveled and the average number of collisions per trial are recorded. This complementary setup provides additional insights into the algorithm's ability to sustain long-term safe and efficient navigation. To ensure the reliability of the results, each method is tested over 200 trials for both experimental setups.

To ensure the reliability of the results, each method is tested multiple times (200 trials for each method).In each trial, the AV starts from rest at a predetermined starting point and accelerates to a target speed of 8 m/s. The experimental results, as shown in Table III, clearly demonstrate that the proposed method significantly outperforms other approaches across various performance metrics in both the infinite mileage and fixed-time tests. The baseline method achieves the highest driving distance in the fixed-time test but suffers from a high collision rate due to its inability to handle complex occlusion scenarios safely. The GO-POMDP method strikes a balance between distance and safety but remains inferior to the proposed method in both metrics. The set-based method, while achieving the lowest collision rate, fails to demonstrate scalability and efficiency due to its tendency to frequently fall into deadlock situations. The proposed method achieves the highest average driving distance in the infinite mileage test and maintains the longest average simulation time of 156.4 seconds, indicating its superior ability to avoid collisions and deadlocks. In the Fixed-Time Test, the proposed method demonstrates an exceptional balance between safety and efficiency.

TABLE III

PERFORMANCE EVALUATION IN INFINITE MILEAGE AND FIXED-TIME TESTS

Method	Infinite Mileage Test		Fixed-Time Test	
	Average Driving Distance (m)	Average Simulation Time (s)	Average Driving Distance (m)	Average Number of Collisions (times)
Baseline	113.8	32.4	626.4	11.380
GO-POMDP	143.5	53.2	538.7	4.975
Set-based method	32.7	19.5	36.8	0
Proposed method	478.1	156.4	572.9	0.315

The proposed method showcases its ability to handle complex occluded environments and challenging scenarios effectively. It achieves the best overall performance, with a significant reduction in collisions, efficient navigation, and robust handling of potential risks, making it a highly reliable and scalable approach for autonomous driving in real-world scenarios.

VI. CONCLUSION

In this paper, we propose an occlusion-aware trajectory planning method aimed at ensuring safe autonomous driving through occluded scenarios. The proposed approach is distinguished by three key features. Firstly, it accounts for potential obstacles within blind spots in both lateral and longitudinal planning, emulating human-like driving behavior when navigating occluded regions. Secondly, we incorporate the concept of phantom obstacle (PHO) and utilize Stochastic Model Predictive Control (SMPC) to account for the significant uncertainties associated with occluded areas and potential obstructed obstacles, ensuring consideration of worst-case scenarios. Additionally, to avoid overly conservative planning outcomes, we further analyze the causes of deadlock on the ST projection and introduce quantified risk constraints. These constraints encourage autonomous vehicles to proceed, explore occluded areas, and facilitate speed decision transitions. Experimental results from diverse occlusion scenarios, encompassing individual cases and statistical analyses, affirm the advantageous impact of lateral trajectory deviation for efficient blind-spot traversal. Different risk factors effectively modulate the conservatism of the planning trajectory in response to occluded regions. Furthermore, the proposed quantified risk constraints prove effective in resolving deadlock situations, promoting forward motion in autonomous vehicles. Finally, owing to comprehensive considerations of various uncertainties, our proposed approach not only enhances passage efficiency but also maintains a lower collision probability compared to the other methods, underscoring the generalizability, robustness, efficiency, and safety of the proposed method. In the future, it would be interesting to validate our method on more diverse testing platforms and evaluate it across a wider

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range of traffic scenarios involving blind spots. Additionally, we aim to improve the algorithm's adaptability to a broader set of environments, enhancing its real-world applicability.

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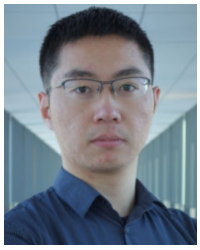

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